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# PEARL MILLET GRAIN HYBRIDS

*by*
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Pearl millet as a short-statured, grain-type, hybrid crop has made a great deal of progress in recent years, with a major plant breeding effort by Mr. William Stegmeier at the Fort Hays Branch Experiment Station of Kansas State University. Experimental hybrids created were evaluated in 1992 at Garden City under dryland field conditions and compared with a grain sorghum hybrid check (DeKalb DK39Y). Resulting data are given in Table 1.

Many of the pearl millet hybrids yielded significantly more grain then did the sorghum hybrid. The test was seeded with three replications on June 16, 1992. The pearl millets averaged 26% greater grain yield than DK39Y sorghum. Additionally, many of the millets had quicker maturity and higher test weights then did the sorghum. This type of pearl millet is becoming a competitive feed grain crop for this region. Additional stalk strength is needed to reduce lodging, but excellent gains have been make in seed size, emergence vigor, and seed set with cool night temperature, along with desired plant and grain yields.

**Table 1. Pearl millet hybrid results in 1992 as compared with sorghum check.**

| Hybrid I.D.# | Height Index | Days to Bloom | % Lodging | Test Wt. | Grain lbs/acre |
|---|---|---|---|---|---|
| 92-105 | 45 | 53 | 50 | 59 | 3332 |
| 92-112 | 43 | 55 | 3 | 60 | 3676 |
| 92-115 | 43 | 55 | 17 | 60 | 3831 |
| 92-125 | 55 | 66 | 17 | 60 | 3896 |
| 92-126 | 56 | 67 | 13 | 60 | 4585 |
| 92-403 | 49 | 54 | 3 | 59 | 3333 |
| 92-412 | 54 | 61 | 82 | 60 | 3280 |
| 92-416 | 57 | 68 | 68 | 59 | 4372 |
| 92-424 | 55 | 63 | 33 | 60 | 4378 |
| 92-437 | 52 | 59 | 32 | 61 | 3985 |
| 92-438 | 47 | 56 | 32 | 59 | 3707 |
| 92-439 | 45 | 61 | 7 | 60 | 3329 |
| 91-3542 | 55 | 59 | 20 | 61 | 3487 |
| 91-3548 | 54 | 57 | 33 | 59 | 3416 |
| 91-3552 | 53 | 58 | 37 | 61 | 3903 |
| 91-3555 | 54 | 58 | 63 | 59 | 3327 |
| 91-3556 | 59 | 58 | 23 | 58 | 3000 |
| 91-3558 | 53 | 58 | 50 | 60 | 3890 |
| 91-3564 | 58 | 60 | 45 | 58 | 3804 |
| 91-3567 | 59 | 63 | 70 | 60 | 4436 |
| 91-3574 | 56 | 54 | 37 | 58 | 3058 |
| 91-3579 | 52 | 57 | 40 | 59 | 3643 |
| 91-3604 | 66 | 70 | 70 | 60 | 3951 |
| 91-3619 | 46 | 56 | 60 | 59 | 4140 |
| 91-3641 | 57 | 66 | 73 | 61 | 4291 |
| 91-3647 | 48 | 55 | 28 | 60 | 3287 |
| 91-3673 | 50 | 57 | 52 | 61 | 3341 |
| 92-0021 | 48 | 56 | 10 | 60 | 2680 |
| 92-0141 | 49 | 57 | 50 | 60 | 3200 |
| DK39Y sorghum (Check) | 45 | 65 | 19 | 56 | 2901 |
| Test Av. | 52 | 59 | 38 | 59 | 3649 |
| L.S.D. (.05) | 4.8 | 2.5 | 36.9 | 2.8 | 621 |